

Turtle Games: Predictive Analysis



Context

Turtle Games (TG) is an international games manufacturer seeking to improve sales. With Loyalty Points earned with each purchase, Loyalty Points reflect sales and the TG customer base. The aim of the TG team is to refine their understanding of Loyalty Points and thus their customer base, to maximise sales. We've used advanced data analysis to investigate the makeup of customers loyalty points, the segmentation of customers for marketing and applied text data analysis too. Furthermore, predictive models will provide insight into the loyalty point scheme.

Analytical Approach

We start by loading the 'turtle_reviews.csv' dataset into python to clean and explore. We identify zero null values, and notable exploratory analyses include mean 'Age' and 'Remuneration' of 39 and £48k respectively. Also, mean 'Loyalty Points' are 1578. For simplicity and the extent of the analysis we rename 'spending_score (1-100)' and 'remuneration (k£)' to remove the bracketed content. All sales are from the web and in EN language so we drop these variables.

Beyond 'numpy', 'pandas' and 'matplotlib' libraries, python libraries key to our analysis include: 'statsmodels.api' and 'statsmodels.stats.api' specifically targeting statistics. 'sklearn' is imported for to allow us to call additional Machine Learning tools such as regression models ('linear_model', 'DecisionTreeRegressor', 'k_means') and analysis tools ('variance_inflation_factor', 'accuracy_score'). For our Natural Language Processing (NLP), we import 'nltk' and also import wordcloud, tokenizing and phrase extraction packages ('wordcloud', 'word_tokenize', 'stopwords', 'textblob').

We initiate analysis with a single linear regression of each spending, remuneration and age separately against loyalty points. The important function here is creating the regression function itself:

To achieve:

$$y = mx + b$$

Use this python code to create the model:

```
lin_spending = LinearRegression()  
lin_spending.fit(x,y)
```

And this scatter graph to visualise it:

```
plt.scatter(x, y, color='blue', label='Given Data')  
plt.plot(x, predicted_loyalty_s, color='red', label='Regression line')
```

Where 'predicted_loyalty_s' points to 's' for spending etc.

We compare the outputs of these regressions. To measure the regressions trueness of fit with R-squared scores, parameters and coefficients we use the following functions:

```
print("R-squared: ", lin_spending.score(x, y))
print("Intercept:", lin_spending.intercept_)
print("Coefficients:")
list(zip(x, lin_spending.coef_))
```

Next we draw conclusions from decision trees where important functions include splitting data into 'test' and 'train' to account for overfitting (A) and creating the decision tree regressor (B). Notably, we adjusted the pruning of the decision tree several times to improve the eligibility and potential over fitting by decreasing the max depth to '3'.

To perform K means analysis, important functions include the elbow and silhouette methods (C), whereby we identified 5 as the optimal cluster number. This, we applied to our final model.

Our last instance on python is to apply NLP to the 'review' and 'summary' variables. The most important code was to determine the polarity and sentiment of the text:

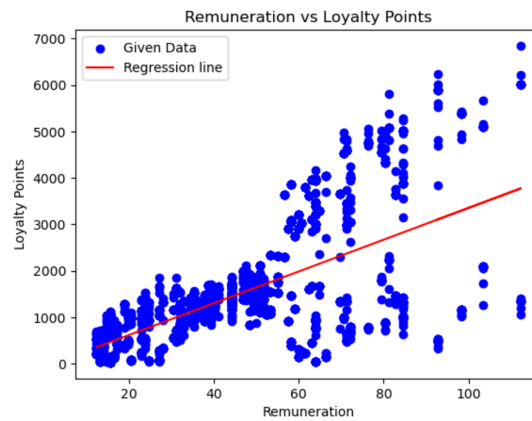
```
df_tokenise['review_polarity'] =
df_tokenise['review'].astype(str).apply(generate_polarity)

df_tokenise['summary_polarity'] =
df_tokenise['summary'].astype(str).apply(generate_polarity)
```

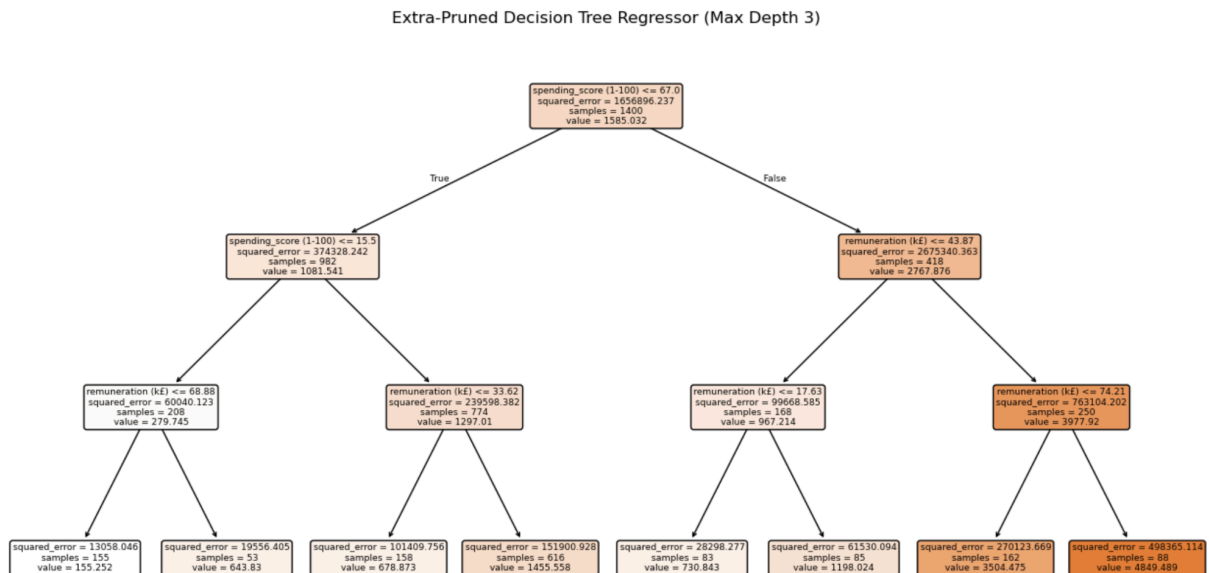
Analysis: Linear Regression, MLR, Decision Tree, NLP

Within the platform 'R', using the same dataset, it was important to install packages for exploring (summarytools) and visualising (gplot2). To perform Multiple Linear Regression, we start by cleaning the data through renaming columns as we did in python. Then summarise (D) and explore data (E). We found there were slightly more females than males (56% vs 44%). From here we produced histogram, boxplot, scatterplot and correlation matrix (F). Notably Loyalty Points were slightly skewed to the right.

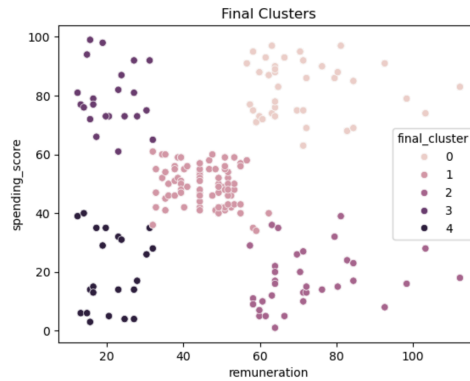
Onto visualising the data - with the linear regression model, we measure the relationship of each independent variable - spending, remuneration and age - against the dependent variable 'Loyalty Points' in isolation. We start by fitting 'Age' linearly - however with evidence of scatter plot (G), we inverse square age to create a new line of best fit. This improved R-squared and hence goodness of fit from 0.0018 to 0.014. A very poorly fit model but still an improvement.



Remuneration and spending are better fitting models, with R-Squared values of 0.38 and 0.45 respectively. The highest coefficient is remuneration whereby for each extra thousand pounds earned, an observation is predicted to increase Loyalty Points by 34, compared to 33, meaning a slightly steeper regression line.



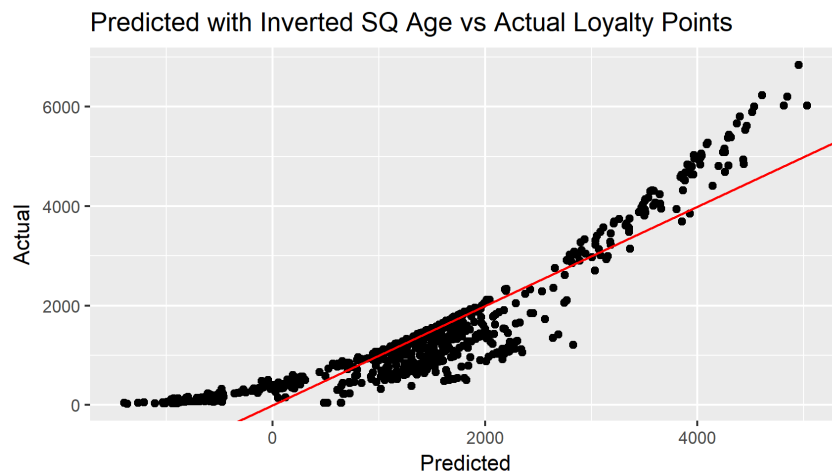
We built three decision trees with increased pruning to improve clarity and avoid overfitting. The full tree had an R-squared of 0.994 - likely overfitting. The final pruned decision tree has a max depth of 3 with an R-squared of 0.914. This tree clearly shows that increasing spending and remuneration in customers is significant in increasing loyalty points.



Regarding K-means, having identified 5 as the optimum number of clusters - we apply our final scattergraph whereby cluster '0' is identified as our most valuable customer base - high spend and high remuneration. Whilst cluster 1 is our largest, with 774 observations, 39% per cent of the total group. These will be our two most valuable customer sets to target in marketing.



Finally, using python, we retrieve wordclouds from the tokenised text in 'summary' and 'reviews', that also have stopwords removed. These clearly show that 'Nan' and words with positive sentiments such as 'Love' and 'Great' are the most frequent. We then create histograms of the polarity of the two variables - that shows both are more positively skewed - with 'summary' being largely neutral (H).



Within R - we applied the inverse square of age to adapt our MLR model later to improve fit. We visualise both the regression models, with a line of best fit. There is little to no distinction between the two regression scatterplots, however both show a positive correlation between predicted and actual values. The spread increases at higher values meaning predictions are more accurate at central values.

Patterns, Trends & Insights

In conclusion, to maximise Loyalty Points and thus sales, TG should focus on customers who already accumulate and are predicted to accumulate loyalty points - those with high income and high remuneration - focussing less on age. Through linear regression analysis, we identified that a customer's age has the smallest impact, whilst remuneration and spending have an almost equal strong positive correlation with Loyalty Points. This is further supported by the coefficients of income and spending in the Multiple Linear Regression model - both 34. Hence to maximise loyalty points, income and spending of customers should be targeted.

The segmentation of customers for the TG marketing team to focus on also follows suit. Through K-means clustering we identified the high income and high spending as a strong cluster - also that the largest and likely most valuable portion of the customer base are the mid income and spending. Specifically, via NLP we identified positive sentiments and consistent reference to 'Nan' in reviews. With this, TG should market the games towards families where older generations can still participate.

Finally, the next steps in the analysis will be to see if the Multiple Linear Regression can be better fit, testing more non-linear fitting to further improve the model and improve predictions.

Appendices

(A) Splitting data into test and train + fitting to model:

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

```
X_train['gender'] = X_train['gender'].map({'Male': 0, 'Female': 1})
X_train['education'] = X_train['education'].astype('category').cat.codes
X_train['platform'] = X_train['platform'].astype('category').cat.codes
```

```
X_test['gender'] = X_test['gender'].map({'Male': 0, 'Female': 1})
X_test['education'] = X_test['education'].astype('category').cat.codes
X_test['platform'] = X_test['platform'].astype('category').cat.codes
```

(B) Create the decision tree regressor

```
from sklearn.tree import DecisionTreeRegressor
regressor = DecisionTreeRegressor(random_state=42)
regressor.fit(X_train, y_train)
```

(C) Elbow and Silhouette Function

Elbow

```
sse = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(df3)
    sse.append(kmeans.inertia_)
```

```
plt.plot(range(1, 11), sse, marker='o')
plt.xlabel('Number of clusters')
plt.ylabel('SSE (Inertia)')
plt.title('Elbow Method')
plt.show()
```

Silhouette:

```
for k in range(2, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    labels = kmeans.fit_predict(df3)
    score = silhouette_score(df3, labels)
    print(f'Silhouette score for k={k}: {score:.3f}')
```

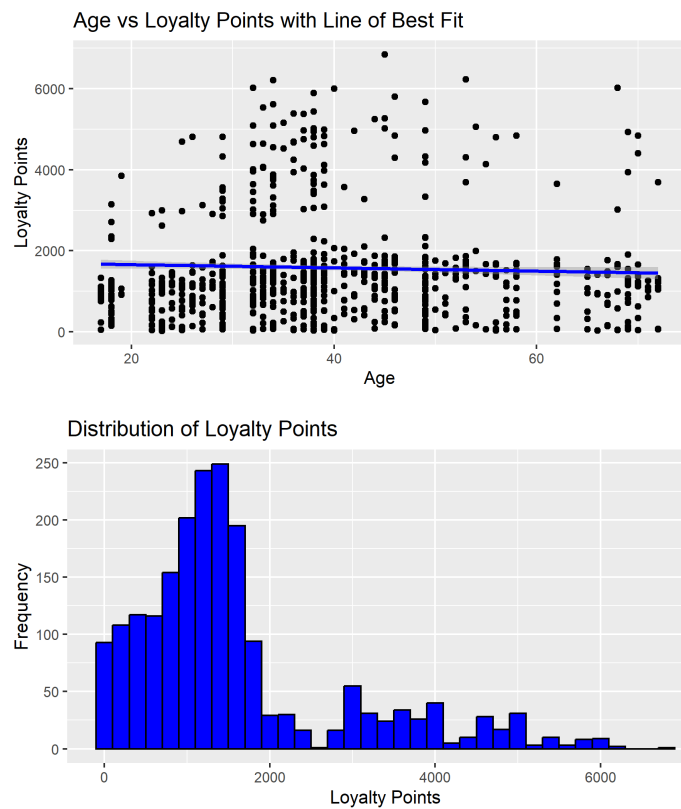
(D) R - Summary

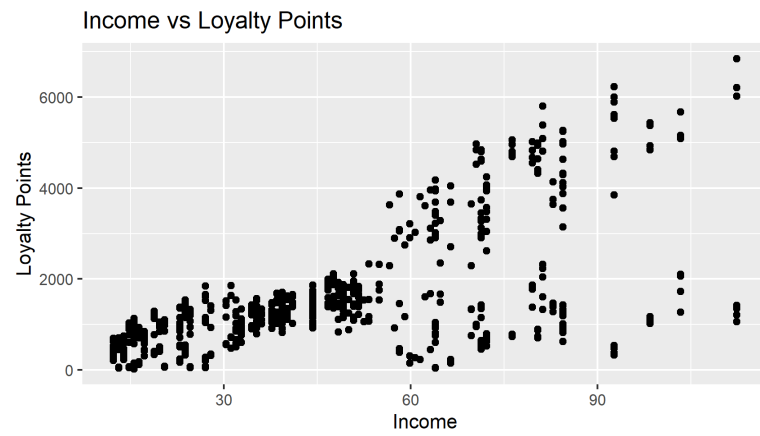
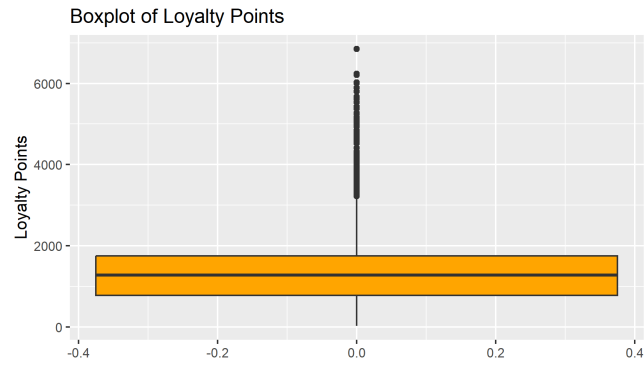
```
head(df)
str(df)
summary(df)
dfSummary(df)
```

(E) R - Explore

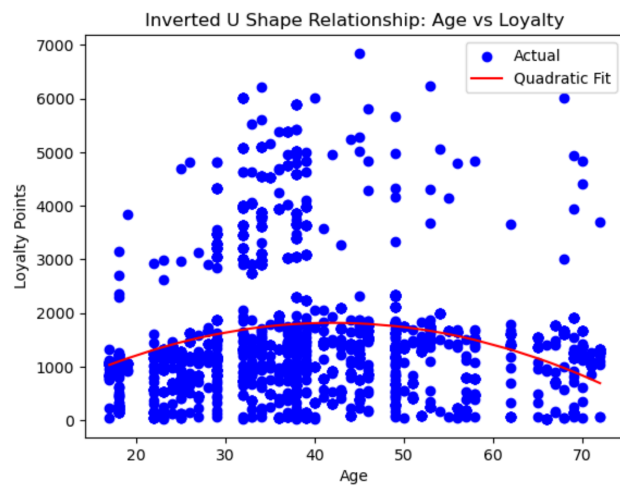
```
colSums(is.na(df))
summary(df)
```

(F) Histogram, boxplot, scatterplot and correlation matrix





(G) Inverse Square Age Scatterplot



(I) Text Polarity

